

EEG x Deep Learning research assignment

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All numbers come from `python main.py` (fixed seeds). Every run is in the files named in brackets. Choices are in `DECISIONS.md` (D-numbers), predictions and their scores in `PREDICTIONS.md`.

Task 1: Motor imagery (EEGBCI, subjects 70-109)

A1 Data audit (`audit.csv`)

Runs	4, 8, 12: imagined left vs right fist (T1 = left, T2 = right). Verified: MNE <code>eegbci</code> docstring, PhysioNet page (D1)
Anomalies	S88, S92, S100 have 128 Hz headers. Header is correct (no 0.8x alpha shift), so kept and resampled to 160 Hz (D2)
Dropped	20 trials whose -1.5 to 4.0 s window runs past the recording, listed per subject (D3)
Kept	897 left, 896 right trials

A2 Physiological ground truth (`a2_lateralisation.csv`, `outputs/erd/`)

Reference	Undocumented, so a lateral reference would bias C3 vs C4. Local Laplacian (C3/C4 minus mean of 4 nearest electrodes) cancels it (D4, D10)
Power	Morlet, baseline -1.0 to -0.1 s, window 0.5-4.0 s (D5)
Index	$LI = (ipsi - contra) / (ipsi + contra)$, μ and β
Test	One-sided label permutation, 10 000 permutations, Bonferroni over 2 bands. Present only if contralateral power drops (D6)
Result	Lateralised ERD present in 14 of 40 subjects. Mean LI 0.26 (μ), 0.32 (β). CAR 0.19/0.29, raw 0.15/0.21

A3 Confound audit (`a3_summary.csv`, `a3_per_subject_table.csv`)

One restricted decoder per non-motor route, each with a pre-cue twin (D12-D17). Threshold: exact one-sided binomial at 5% (52.0% pooled over 1793 trials, 63-67% within subject).

Decoder (route)	Within subject	Above threshold	Pooled	Pre-cue
F1 frontal 0-0.5 s (eye)	66.4%	25/40	71.9%	48.2%
F2 frontal 0.5-4 s	63.5%	18/40	75.3%	-
O1 occipital 0-0.5 s (visual)	56.5%	14/40	59.3%	49.7%
G1 temporal 30-40 Hz (muscle)	50.8%	1/40	49.6%	49.3%
M1 sensorimotor μ + β	54.9%	8/40	54.6%	52.2%*
(motor)				
R2 previous 1-3 labels (no EEG)	70.8%	31/40	72.6%	-

*Marginal (permutation p 0.02-0.03 among 10 uncorrected twins). Label order alternates ($z = -17.6$). A zero-phase filter leaked into the pre-cue window, so all results use a causal filter (D18). Prediction (motor 56-62%, order at chance): wrong.

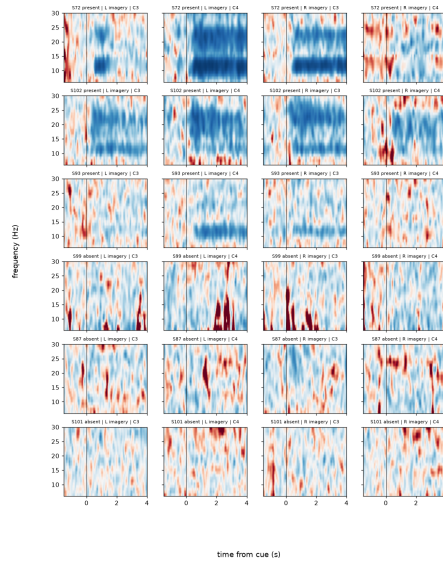


Figure 1: ERD at C3 and C4 relative to baseline (blue = desynchronisation). Top three: present (S72, S102, S93). Bottom three: absent (S99, S87, S101).

B1 Models and evaluation

Model	Parameters	Details
EEGNet-8,2	2962	0.5 s kernels rescaled 128 to 160 Hz: 64 to 80 and 16 to 20 samples (D20)
Transformer, time-patch	6834	d_model 16, 2 layers, 2 heads, EEGNet temporal filter (D21)
Transformer, channel + identity	7250	learned electrode identity
Transformer, channel, no identity	6226	

Cross-subject only: 8 fixed subject-wise folds of 5, fold f+1 for early stopping (D22). B2 tests on folds 0-3 (20 subjects, 913 trials, threshold 52.8%), seeds 0-2 (D24).

B2 Experiments (outputs/partb/b2_all_runs.csv, B2_RESULTS.md)

Accuracy %, mean $\pm$ SD (3 seeds)	5 subj.	10	20	30	30, eye route removed
EEGNet	72.1 $\pm$ 0.5	75.0 $\pm$ 1.3	76.2 $\pm$ 1.3	76.9 $\pm$ 1.0	50.5 $\pm$ 2.9
Transformer, time-patch	54.8 $\pm$ 3.1	64.3 $\pm$ 3.0	66.8 $\pm$ 4.1	71.6 $\pm$ 3.2	49.1 $\pm$ 1.3
Transformer, channel + identity				70.2 $\pm$ 1.2	
Transformer, channel, no identity				57.5 $\pm$ 1.6	

Experiment	Result
Displacement (test time, spline interpolation)	< 1 point lost at 10 mm, $\leq$ 1.5 at 20 mm. Time-patch transformer drops most
Faithfulness (delete half the tokens)	Most-attended 71.4% to 57.7%, gradient x input 64.1%, random 65.6%
What it relies on	F7, F8, Ft7, Ft8, Af8 (horizontal EOG sites), the A3 eye route
Link to Part A (Spearman, per test subject)	EEGNet vs A2 LI 0.72, vs A3 frontal decoder 0.53. LI vs frontal -0.02
Confound removed (Laplacian, 21 Fc/C/Cp channels, D25)	EEGNet 50.5%, transformer 49.1%: chance

Experiment	Result
Intervention (random 0-15 mm slides in training, D26)	No effect: 77.7% vs 76.9%

Write-up

(a) If a model scores 70% on these subjects, how much needs motor imagery?

Almost none.

- Eye route alone (frontal, 0-0.5 s after the cue): 71.9% pooled, above threshold in 25/40. Its pre-cue twin is at chance (48.2%).
- Motor route (sensorimotor μ +beta): 54.6% pooled, 8/40, although A2 finds lateralised ERD in 14/40. The motor signal is real but differs between subjects. The eye signal has the same sign in everyone.
- Both networks rely on F7, F8, Ft7, Ft8, Af8 (horizontal EOG sites) and fall to chance without the eye route (EEGNet 50.5%, transformer 49.1%).

(b) B2 mechanisms

Result	Mechanism
No crossover with data (predicted a crossover: wrong)	EEGNet hard-codes a filter bank with per-band spatial filters, so ~2900 weights fit the broad frontal eye pattern from 5 subjects. The transformer must learn how 20 time segments combine, which costs data
Channel tokens without identity: 57.5%	Attention with a CLS read-out is permutation-invariant over electrodes, so C3 and C4 look the same
Displacement costs < 1 point at 10 mm	EEGNet's temporal pooling averages the spatial perturbation down (17% to 13%). The transformer's spatial filters feed every token, so it drops most
Attention beats gradient in faithfulness	Deleting tokens renormalises the softmax, which last-layer attention captures and a first-order gradient does not
Confound removed: chance (predicted a small drop: wrong)	The motor pattern is subject-specific, so it does not transfer. Only the eye pattern is shared across subjects
Intervention: no effect (predicted)	Training on slides penalises the loss change along a slide, which displacement showed is already negligible

Task 2: Sleep staging (Sleep-EDF, subjects 0-14)

2a Baseline (task2/baseline/2a_baseline_seed42.txt)

Given script, unchanged, seed 42: **accuracy 0.847, macro-F1 0.628, kappa 0.693**. N1 F1 0.207. 70% of test windows are Wake.

2b Defects (DEFECTS.md, task2/evidence/, one commit each)

#	Defect	Evidence	Fix
1	split by recording	5/5 test subjects' other night in training	split by subject, 9/3/3
2	epoch chosen on test	test F1 0.58-0.67 over 12 epochs, max reported	select on validation
3	batch_first=False	own neighbour changes output by 0, other window by 9e-4	batch_first=True
4	position after attention	shuffled neighbours change output by 3e-7	encode before
5	label = last epoch	differs from centre in 8.7% of windows, 71% of N1	label centre epoch
6	500 "uV" on volts	0 of 80070 rejected, units-only fix rejects 54% (EOG)	uV, EEG only
7	rejected epochs deleted	windows span time gaps	mask, skip windows
8	per-epoch z-score	99.6% of N3 > 75 uV, all std = 1 after	per-recording z
9	unscored as Wake	905 "?" + 35 movement epochs	label -1
10	~23 h recordings	68% lights-on Wake, "always Wake" = 68.7%	crop to sleep ± 30 min

Fixed pipeline, seeds 42-44: **accuracy 0.827 ± 0.012, macro-F1 0.788 ± 0.015, kappa 0.769 ± 0.020**. Judgement calls: T2-1 to T2-10.

2c Impact ledger (task2/ledger.py, task2/ledger_table.md)

Each defect re-introduced alone into the fixed script. Δ = defective - fixed, same seed, mean ± SD over seeds 42-44 († seeds 42-43 only).

Defect	Δ accuracy	Δ macro-F1	Δ kappa	Predicted (F1)	Outcome
10 no crop	+0.098 ± 0.017	+0.013 ± 0.018	+0.089 ± 0.028	acc up, F1 ~0	right
3 batch attention	-0.026 ± 0.022	-0.034 ± 0.020	-0.035 ± 0.027	down	right
8 per-epoch z	-0.012 ± 0.010	-0.025 ± 0.019	-0.020 ± 0.013	down, N3 most	N1 most
9 unscored as W	-0.015 ± 0.007	-0.017 ± 0.007	-0.020 ± 0.006	~0	larger
2 select on test	+0.010 ± 0.009	+0.016 ± 0.014	+0.015 ± 0.013	up	right
1 recording split	-0.007 ± 0.023	-0.012 ± 0.008	-0.014 ± 0.036	up	wrong sign
4 position after	-0.003 ± 0.021	-0.006 ± 0.023	-0.003 ± 0.027	small down	noise
5 last-epoch label	+0.009 ± 0.011	+0.007 ± 0.015	+0.013 ± 0.014	down	noise
6 units	+0.006 ± 0.010	+0.008 ± 0.014	+0.009 ± 0.016	~0	right (noise)
7 deletion	+0.011 ± 0.019	+0.008 ± 0.015	+0.014 ± 0.024	~0	right (noise)
6N naive units fix †	-0.028 ± 0.004	-0.049 ± 0.002	-0.069 ± 0.000	down	right
6N + 7 †	-0.016 ± 0.015	-0.021 ± 0.026	-0.029 ± 0.024	worse than 6N	flips, less bad
3 + 4	-0.037 ± 0.016	-0.046 ± 0.031	-0.048 ± 0.024	= 3	right
1 + 2	+0.015 ± 0.033	+0.012 ± 0.026	+0.014 ± 0.053	more than either	not additive
9 + 10 †	+0.097 ± 0.012	+0.005 ± 0.017	+0.081 ± 0.021	9 grows	9 vanishes
all ten †	+0.027 ± 0.031	-0.147 ± 0.032	-0.059 ± 0.054	acc up, F1 -0.08..-0.20	right

Noise floor. Runs are bit-identical at the same seed and thread count. Changing 11 of ~17,000 training windows moves macro-F1 by 0.024, so effects under ±0.02 are not resolved.

Ranked by how much each would have misled the report: 10 (inflates accuracy and kappa through lights-on Wake), 3 (the context model never ran), 8 (removes the N3 amplitude rule), then 9 and 2. 1, 4, 5, 6, 7 are within noise.

Flags. 4 vanishes under 3. 7 vanishes under 6 and flips with the naive units fix (masking keeps 11k training windows, deletion 17.3k). 9 vanishes once Wake is uncropped. 1 has the wrong sign because the seed also draws a harder test set.

2d Worst stage (task2/PHYSIOLOGY.md)

	W	N1	N2	N3	REM
F1 (3 seeds)	0.889	0.511	0.851	0.862	0.827
Chin EMG median (uV)	3.23	2.68	1.98	1.48	0.69

Signals: EEG Fpz-Cz, Pz-Oz and EOG at 100 Hz. Chin EMG only as a 1 Hz envelope, not used by the pipeline.

- Worst: **N1** (precision 0.44, recall 0.61). On EEG and EOG it matches REM (Fpz-Cz amplitude 78 vs 79 uV). Only muscle tone separates them.
- Expected confusion: N3 scored as N2 (17.5%). The 20% slow-wave rule cuts a continuum.
- Unexpected confusion: W and REM (5.1%, 3.3%). Chin EMG separates them, but it is not in the input.

2e Note for a non-technical reader

Our first sleep-staging program reported 85% accuracy. The corrected version scores about 83%, and the lower figure is the honest one.

The first test was too easy. Each recording runs for almost a whole day, and for two-thirds of it the person is awake. A program that always answered “awake” would score 69%. The corrected test keeps only the night, plus half an hour either side.

The old test had two other flaws. It included people the program had already studied, since their other night was in its training material. And its best version was picked using the test itself. Real patients will be new to it.

A fairer score gives every sleep stage equal weight, so easy wakefulness earns nothing extra. On that score the corrected program improved from 0.63 to 0.79 (out of 1). The headline dropped because the test got harder, while the program got better at telling sleep stages apart.